THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 5: **Circles** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 90°	An angle in a semicircle is a right angle.
2	**(a)** 7	Radius $= \frac{d}{2}$.
3	**(b)** 5	The perpendicular from the centre bisects the chord.
4	**(c)** 25°	The angle at the centre is double the angle at the circumference.
5	**(c)** 40°	Central angle $= 2\times$ angle at the circumference.
6	**(d)** 115°	Opposite angles of a cyclic quadrilateral are supplementary.
7	**(b)** diameter	Recall the circle theorems.
8	**(a)** 90°	An angle in a semicircle is a right angle.
9	**(b)** 9	Radius $= \frac{d}{2}$.
10	**(d)** 8	The perpendicular from the centre bisects the chord.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Angle at the centre is double that at the circumference (1); Required angle $= \frac{36^\circ}{2} = 18^\circ$ (1).	1+1
12	Opposite angles are supplementary (1); $\angle D = 180^\circ - 72^\circ = 108^\circ$ (1).	1+1
13	$\angle ACB = 90^\circ$ (angle in a semicircle) (1); $\angle ABC = 180^\circ - 90^\circ - 28^\circ = 62^\circ$ (1).	1+1
14	Angle at the centre is double that at the circumference (1); Required angle $= \frac{84^\circ}{2} = 42^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Half-chord $= \sqrt{5^2 - 3^2}$ (1); $= \sqrt{16} = 4$ cm (1); Chord $= 2 \times 4 = 8$ cm (1).	1+1+1
16	Angle at the centre is double that at the circumference (1); $\angle ACB = 24^\circ$ (1); Angles in the same segment are equal (1).	1+1+1
17	Opposite angles are supplementary (1); $\angle C = 180^\circ - 70^\circ = 110^\circ$ (1); $\angle D = 180^\circ - 85^\circ = 95^\circ$ (1).	1+1+1
	OR: Half-chord $= \sqrt{5^2 - 4^2}$ (1); $= \sqrt{9} = 3$ cm (1); Chord $= 2 \times 3 = 6$ cm (1).	1+1+1
18	Angle at the centre is double that at the circumference (1); $\angle ACB = 33^\circ$ (1); Angles in the same segment are equal (1).	1+1+1
	OR: Opposite angles are supplementary (1); $\angle C = 180^\circ - 100^\circ = 80^\circ$ (1); $\angle D = 180^\circ - 75^\circ = 105^\circ$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	The perpendicular from the centre bisects the chord (1); Half-chord $= 12$ cm (1); $r = \sqrt{12^2 + 9^2} = \sqrt{225}$ (1); $r = 15$ cm (1); Radius found using the right triangle (1).	1+1+1+1+1
	OR: Angle in a semicircle is 90° so $\angle ACB = \angle ADB = 90^\circ$ (1); $\angle ABC = 90^\circ - 25^\circ = 65^\circ$ (1); $\angle ABD = 90^\circ - 30^\circ = 60^\circ$ (1); $\angle CBD = 125^\circ$ is not needed here (1); Sum check: $\angle ABC + \angle ABD = 125^\circ$ (1).	1+1+1+1+1
20	The perpendicular from the centre bisects the chord (1); Half-chord $= 9$ cm (1); $r = \sqrt{9^2 + 12^2} = \sqrt{225}$ (1); $r = 15$ cm (1); Radius found using the right triangle (1).	1+1+1+1+1
	OR: Angle in a semicircle is 90° so $\angle ACB = \angle ADB = 90^\circ$ (1); $\angle ABC = 90^\circ - 35^\circ = 55^\circ$ (1); $\angle ABD = 90^\circ - 40^\circ = 50^\circ$ (1); $\angle CBD = 105^\circ$ is not needed here (1); Sum check: $\angle ABC + \angle ABD = 105^\circ$ (1).	1+1+1+1+1

Mistakes to avoid:

- Using the full chord instead of the half-chord in $r^2 = d^2 + (\text{half-chord})^2$.
- Forgetting that the perpendicular distance, not the slant distance, is used when finding the distance of a chord from the centre.
- Applying the angle theorems to an arc without checking the angle is at the *centre* (double) or on the *remaining* circle (half).
- For two parallel chords, adding the distances when they are on the same side or subtracting them when they are on opposite sides (the reverse is correct).
- Assuming that opposite angles of a cyclic quadrilateral are equal; they are supplementary (sum 180°).

- Using the angle in a semicircle theorem when the chord is not a diameter.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 5: **Circles** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 32°	The angle at the centre is double the angle at the circumference.
2	**(b)** 70°	Central angle $= 2\times$ angle at the circumference.
3	**(a)** 100°	Opposite angles of a cyclic quadrilateral are supplementary.
4	**(d)** exactly one	Recall the circle theorems.
5	**(d)** 90°	An angle in a semicircle is a right angle.
6	**(a)** 12	Radius $= \frac{d}{2}$.
7	**(c)** 9	The perpendicular from the centre bisects the chord.
8	**(c)** 40°	The angle at the centre is double the angle at the circumference.
9	**(a)** 96°	Central angle $= 2\times$ angle at the circumference.
10	**(c)** 85°	Opposite angles of a cyclic quadrilateral are supplementary.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Opposite angles are supplementary (1); $\angle D = 180^\circ - 88^\circ = 92^\circ$ (1).	1+1
12	$\angle ACB = 90^\circ$ (angle in a semicircle) (1); $\angle ABC = 180^\circ - 90^\circ - 37^\circ = 53^\circ$ (1).	1+1
13	Angle at the centre is double that at the circumference (1); Required angle $= \frac{108^\circ}{2} = 54^\circ$ (1).	1+1
14	Opposite angles are supplementary (1); $\angle D = 180^\circ - 105^\circ = 75^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Half-chord $= \sqrt{10^2 - 6^2}$ (1); $= \sqrt{64} = 8$ cm (1); Chord $= 2 \times 8 = 16$ cm (1).	1+1+1
16	Angle at the centre is double that at the circumference (1); $\angle ACB = 41^\circ$ (1); Angles in the same segment are equal (1).	1+1+1
17	Opposite angles are supplementary (1); $\angle C = 180^\circ - 64^\circ = 116^\circ$ (1); $\angle D = 180^\circ - 92^\circ = 88^\circ$ (1).	1+1+1
	OR: Half-chord $= \sqrt{10^2 - 8^2}$ (1); $= \sqrt{36} = 6$ cm (1); Chord $= 2 \times 6 = 12$ cm (1).	1+1+1
18	Angle at the centre is double that at the circumference (1); $\angle ACB = 52^\circ$ (1); Angles in the same segment are equal (1).	1+1+1
	OR: Opposite angles are supplementary (1); $\angle C = 180^\circ - 118^\circ = 62^\circ$ (1); $\angle D = 180^\circ - 83^\circ = 97^\circ$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	The perpendicular from the centre bisects the chord (1); Half-chord $= 15$ cm (1); $r = \sqrt{15^2 + 8^2} = \sqrt{289}$ (1); $r = 17$ cm (1); Radius found using the right triangle (1).	1+1+1+1+1
	OR: Angle in a semicircle is 90° so $\angle ACB = \angle ADB = 90^\circ$ (1); $\angle ABC = 90^\circ - 20^\circ = 70^\circ$ (1); $\angle ABD = 90^\circ - 45^\circ = 45^\circ$ (1); $\angle CBD = 115^\circ$ is not needed here (1); Sum check: $\angle ABC + \angle ABD = 115^\circ$ (1).	1+1+1+1+1
20	The perpendicular from the centre bisects the chord (1); Half-chord $= 8$ cm (1); $r = \sqrt{8^2 + 15^2} = \sqrt{289}$ (1); $r = 17$ cm (1); Radius found using the right triangle (1).	1+1+1+1+1
	OR: Angle in a semicircle is 90° so $\angle ACB = \angle ADB = 90^\circ$ (1); $\angle ABC = 90^\circ - 32^\circ = 58^\circ$ (1); $\angle ABD = 90^\circ - 28^\circ = 62^\circ$ (1); $\angle CBD = 120^\circ$ is not needed here (1); Sum check: $\angle ABC + \angle ABD = 120^\circ$ (1).	1+1+1+1+1

Mistakes to avoid:

- Using the full chord instead of the half-chord in $r^2 = d^2 + (\text{half-chord})^2$.
- Forgetting that the perpendicular distance, not the slant distance, is used when finding the distance of a chord from the centre.
- Applying the angle theorems to an arc without checking the angle is at the *centre* (double) or on the *remaining* circle (half).
- For two parallel chords, adding the distances when they are on the same side or subtracting them when they are on opposite sides (the reverse is correct).
- Assuming that opposite angles of a cyclic quadrilateral are equal; they are supplementary (sum 180°).

- Using the angle in a semicircle theorem when the chord is not a diameter.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 5: **Circles** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** infinitely many	Recall the circle theorems.
2	**(a)** 90°	An angle in a semicircle is a right angle.
3	**(c)** 15	Radius $= \frac{d}{2}$.
4	**(d)** 11	The perpendicular from the centre bisects the chord.
5	**(a)** 55°	The angle at the centre is double the angle at the circumference.
6	**(d)** 124°	Central angle $= 2\times$ angle at the circumference.
7	**(a)** 70°	Opposite angles of a cyclic quadrilateral are supplementary.
8	**(d)** equal	Recall the circle theorems.
9	**(a)** 18	Radius $= \frac{d}{2}$.
10	**(b)** 14	The perpendicular from the centre bisects the chord.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\angle ACB = 90^\circ$ (angle in a semicircle) (1); $\angle ABC = 180^\circ - 90^\circ - 46^\circ = 44^\circ$ (1).	1+1
12	Angle at the centre is double that at the circumference (1); Required angle $= \frac{132^\circ}{2} = 66^\circ$ (1).	1+1
13	Opposite angles are supplementary (1); $\angle D = 180^\circ - 118^\circ = 62^\circ$ (1).	1+1
14	$\angle ACB = 90^\circ$ (angle in a semicircle) (1); $\angle ABC = 180^\circ - 90^\circ - 53^\circ = 37^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Half-chord $= \sqrt{13^2 - 5^2}$ (1); $= \sqrt{144} = 12$ cm (1); Chord $= 2 \times 12 = 24$ cm (1).	1+1+1
16	Angle at the centre is double that at the circumference (1); $\angle ACB = 58^\circ$ (1); Angles in the same segment are equal (1).	1+1+1
17	Opposite angles are supplementary (1); $\angle C = 180^\circ - 57^\circ = 123^\circ$ (1); $\angle D = 180^\circ - 106^\circ = 74^\circ$ (1).	1+1+1
	OR: Half-chord $= \sqrt{13^2 - 12^2}$ (1); $= \sqrt{25} = 5$ cm (1); Chord $= 2 \times 5 = 10$ cm (1).	1+1+1
18	Angle at the centre is double that at the circumference (1); $\angle ACB = 64^\circ$ (1); Angles in the same segment are equal (1).	1+1+1
	OR: Opposite angles are supplementary (1); $\angle C = 180^\circ - 95^\circ = 85^\circ$ (1); $\angle D = 180^\circ - 68^\circ = 112^\circ$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	The perpendicular from the centre bisects the chord (1); Half-chord $= 16$ cm (1); $r = \sqrt{16^2 + 12^2} = \sqrt{400}$ (1); $r = 20$ cm (1); Radius found using the right triangle (1).	1+1+1+1+1
	OR: Angle in a semicircle is 90° so $\angle ACB = \angle ADB = 90^\circ$ (1); $\angle ABC = 90^\circ - 15^\circ = 75^\circ$ (1); $\angle ABD = 90^\circ - 50^\circ = 40^\circ$ (1); $\angle CBD = 115^\circ$ is not needed here (1); Sum check: $\angle ABC + \angle ABD = 115^\circ$ (1).	1+1+1+1+1
20	The perpendicular from the centre bisects the chord (1); Half-chord $= 12$ cm (1); $r = \sqrt{12^2 + 16^2} = \sqrt{400}$ (1); $r = 20$ cm (1); Radius found using the right triangle (1).	1+1+1+1+1
	OR: Angle in a semicircle is 90° so $\angle ACB = \angle ADB = 90^\circ$ (1); $\angle ABC = 90^\circ - 38^\circ = 52^\circ$ (1); $\angle ABD = 90^\circ - 22^\circ = 68^\circ$ (1); $\angle CBD = 120^\circ$ is not needed here (1); Sum check: $\angle ABC + \angle ABD = 120^\circ$ (1).	1+1+1+1+1

Mistakes to avoid:

- Using the full chord instead of the half-chord in $r^2 = d^2 + (\text{half-chord})^2$.
- Forgetting that the perpendicular distance, not the slant distance, is used when finding the distance of a chord from the centre.
- Applying the angle theorems to an arc without checking the angle is at the *centre* (double) or on the *remaining* circle (half).
- For two parallel chords, adding the distances when they are on the same side or subtracting them when they are on opposite sides (the reverse is correct).
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- Using the angle in a semicircle theorem when the chord is not a diameter.

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